## Review Problems

**1.** a) A computer generates randomly passwords consisting only of lower letters and digits. A password has 5 characters what is the probability to have 2 lower case and 3 digits (in any order)?
b) A computer generates randomly passwords consisting of digits, lower letters and capital letters. A password has 7 characters.
$\triangleright$ What is the probability of randomly guessing such a password?
$\triangleright$ What is the probability to have 2 digits, 3 lower letters and 2 capital letters (in any order)?

**A:** 1a) $C_5^2 \left(\frac{10}{36}\right)^2 \left(\frac{26}{36}\right)^3$, where $36 = 26 + 10$;

1b) $\triangleright$ $\frac{1}{62^7}$, where $62 = 10 + 26 + 26$;

$\triangleright$ $\frac{7}{2!3!2!} \left(\frac{10}{62}\right)^2 \left(\frac{26}{62}\right)^3 \left(\frac{26}{62}\right)^2$

**2.** Let $(X_n)_{n\geq 1}$ be a sequence of random variables such that for each $x \in \mathbb{R}$ and $n \geq 1$ we have

$$P(X_n \leq x) = \begin{cases} 1 - \left(1 - \frac{1}{n}\right)^{nx}, & \text{if } x > 0 \\ 0, & \text{otherwise.} \end{cases}$$

Show that $(X_n)_n$ converges in distribution to $X$, where $X \sim Exp(1)$.
Hint: The density function of an $Exp(1)$ distributed random variable is

$$f(t) = \begin{cases} e^{-x}, & \text{if } 0 < x \\ 0, & \text{if } x \leq 0. \end{cases}$$

**A:** For $x \leq 0$, we have $F_{X_n}(x) = P(X_n \leq x) = F_X(x) = 0$, for each $n \geq 1$. For $x > 0$, we write

$$\lim_{n\to\infty} F_{X_n}(x) = \lim_{n\to\infty} 1 - \left(1 - \frac{1}{n}\right)^{nx} = 1 - \lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^{nx} = 1 - e^{-x} = F_X(x).$$

Thus, we conclude that $X_n \xrightarrow{d} X$.

**3.** Let $(X_n)_n$ be a sequence of independent random variables such that for each $n \in \mathbb{N}^*$ the random variable $X_n$ has the cdf $F_{X_n} : \mathbb{R} \to [0,1]$ defined by

$$F_{X_n}(x) = \begin{cases} 0, & x < 0 \\ nx, & 0 \leq x < \frac{1}{n} \\ 1, & \frac{1}{n} \leq x\,. \end{cases}$$

$\triangleright$ Compute the expectation and variance of $X_n$, $n \geq 1$.
$\triangleright$ Prove that $(X_n)_n$ obeys the strong law of large numbers and write down the limit that expresses this law.

**A:**

$$f_{X_n}(x) = \begin{cases} n, & 0 < x < \frac{1}{n} \\ 0, & \text{otherwise}\,. \end{cases}$$

$\Rightarrow E(X_n) = \frac{1}{2n}$; $V(X_n) = \frac{1}{12n^2}$, $n \geq 1$; $\sum\limits_{n=1}^{\infty} \frac{1}{n^2} V(X_n) = \sum\limits_{n=1}^{\infty} \frac{1}{12n^4} < \infty$ (result from Analysis).
By a theorem from the lecture $\Rightarrow$ $(X_n)_n$ obeys the strong law of large numbers:

$$\frac{1}{n} \sum_{k=1}^{n} \left(X_k - \frac{1}{2k}\right) \xrightarrow{a.s.} 0.$$

**4.** A call center receives many customer inquiries daily. From past data, it is known that a customer service representative successfully resolves a caller's issue on the first attempt 60% of the time. For a particular shift, a representative handles 15 calls.
$\triangleright$ What is the probability that exactly 10 calls are successfully resolved on the first attempt?
$\triangleright$ What is the probability that at least 12 calls are resolved on the first attempt?
$\triangleright$ What is the expected number of resolved calls in this shift?

**A:** Let $X$ = random variable that shows the number of successfully resolved inquiries on the first attempt; $X \sim Bino(15; 0.7)$

$P(X = 10) = C_{15}^{10}(0.6)^{10}(0.4)^5$; $P(X \geq 12) = \sum\limits_{k=12}^{15} C_{15}^{k}(0.6)^k(0.4)^{15-k}$;

$E(X) = \sum\limits_{k=0}^{15} kC_{15}^{k}(0.6)^k(0.4)^{15-k} = 15 \cdot 0.6 = 9.$

**5.** A company has developed an email filtering system to detect spam emails. From previous data:

- 10% of all received emails are actually *spam*.
- If an email is *spam*, the system correctly identifies it 95% of the time.
- If an email is *not spam*, the system mistakenly classifies it as *spam* 5% of the time.

A randomly received email has been *marked as spam* by the system.
What is the probability that the email is actually *spam*, given that the system flagged it as *spam*?

**A:** Consider:
$S$: The email is *spam*.
$\bar{S}$: The email is *not spam*.

$$P(S) = 0.1, \quad P(\bar{S}) = 1 - P(S) = 0.9\,.$$

$M$: The system marks the email as *spam*.
$P(M|S)$ is the conditional probability that given an email that is *spam*, the system correctly classifies it as *spam* (i.e. the probability that the system marks a *spam* correctly).
$P(M|\bar{S})$ is the conditional probability that given an email that is *not spam*, the system mistakenly classifies it as *spam* (i.e. the probability that the system marks a *non spam* incorrectly).

$$P(M|S) = 0.95, \quad P(M|\bar{S}) = 0.05$$

Using the law of total probability:

$$P(M) = P(M|S)P(S) + P(M|\bar{S})P(\bar{S}) = (0.95 \cdot 0.1) + (0.05 \cdot 0.9) = 0.14$$

Bayes' formula implies

$$P(S|M) = \frac{P(M|S)P(S)}{P(M)} = \frac{0.95 \cdot 0.1}{0.14} = \frac{0.095}{0.14} \approx 0.6786\,.$$

Thus, given that the system marks an email as spam, the probability that it is actually spam is approximately 67.86%.

**6.** $X_1$,$X_2$ and $X_3$ are random variables having the same cdf, which is given by $F : \mathbb{R} \to [0,1]$

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{4}, & 0 \le x < 1 \\ \frac{3}{4}, & 1 \le x < 2 \\ 1, & 2 \le x\,. \end{cases}$$

Compute (a) the expectation and variance of $X_1$;
(b) the expectation of the random variable $X_1 + 2X_2 + 3X_3$.
A: Same cdf implies same distribution for all 3 random variables

$$X_1, X_2, X_3 \sim \begin{pmatrix} 0 & 1 & 2 \\ \frac{1}{4} & \frac{2}{4} & \frac{1}{4} \end{pmatrix}$$

$\Rightarrow E(X_1) = 0 \cdot \frac{1}{4} + 1 \cdot \frac{2}{4} + 2 \cdot \frac{1}{4} = 1 \Rightarrow E(X_1) = E(X_2) = E(X_3) = 1$
$\Rightarrow E(X_1 + 2X_2 + 3X_3) = E(X_1) + 2E(X_2) + 3E(X_3) = 6.$